recalculate_anova_components.R

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`recalculate_anova_components` *Recalculate ANOVA Components via Shapley-Allocated Percentages*

Description

Converts a long-format ANOVA results table into per-age percentages for the three unique variance-partitioning components (Abiotic, Associations, Spatial) using a Shapley equal-split allocation of intersection terms. Negative Nagelkerke RB2 values are clamped to zero before any computation.

Usage

```
recalculate_anova_components(data_source)
```

Arguments

`data_source` A non-empty data frame with columns:

age Numeric. Age (cal yr BP) of the time slice.

component Character. Component label. May include any of the seven sjSDM variance-partitioning labels: `"Abiotic"`, `"Associations"`, `"Spatial"`, `"Abiotic&Associatio` `"Abiotic&Spatial"`, `"Associations&Spatial"`, and `"Abiotic&Associations&Spatial"`. Missing intersection labels are treated as zero.

R2_Nagelkerke Numeric. Nagelkerke RB2 value. Values below zero are clamped to 0 internally.

Details

Negative clamping: All `R2_Nagelkerke` values are clamped to $[0, \infty)$ before allocation. Negative values arise from suppressor effects and carry no directional attribution.

Shapley equal-split allocation: Each intersection term is divided equally among its constituent unique components, irrespective of unique-fraction magnitudes. The three adjusted values therefore sum to the total of all seven (clamped) fractions, preserving total explained variance. Using equal splits rather than proportional splits avoids a feedback loop in which a larger unique fraction accumulates more shared variance, further inflating its apparent importance.

Concretely (where missing fractions are treated as 0):

- Abiotic_adj = F_A + F_AB/2 + F_AS/2 + F_ABS/3
- Assoc_adj = F_B + F_AB/2 + F_BS/2 + F_ABS/3
- Spatial_adj = F_S + F_AS/2 + F_BS/2 + F_ABS/3

Value

A tibble with exactly 3 rows per age (one per unique component) and columns `age`, `component`, `R2_Nagelkerke_adjusted`, and `R2_Nagelkerke_percentage`:

R2_Nagelkerke_adjusted Numeric. Shapley-adjusted RB2 for the component: the component's unique fraction plus an equal share of every intersection term that includes it.

R2_Nagelkerke_percentage Numeric. Percentage of total adjusted RB2 explained by this component within the age slice: `R2_adjusted / sum(R2_adjusted) * 100`. `NA_real_` when the age-slice total is zero.

See Also

[extract_anova_fractions()], [aggregate_anova_components()]

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